

An Immersive Layout Framework for Web Design in Virtual Reality

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ABSTRACT

In recent years, many consumer-grade VR devices have emerged, and VR devices have the potential to become the next generation of terminals for visiting websites. Compared to traditional PC and mobile devices, VR moves away from the desktop concept and can show information in a 3D environment. This two-dimensional to three-dimensional transformation means that website layouts, media content, and interactions can make the user experience more immersive. Currently, VR users can visit websites through built-in or third-party browsers, but most website content and UI visuals are consistent with the desktop version instead of the VR version. The website has no separate visual and interactive designs for VR devices. Therefore, this paper analyzes the design elements from three aspects: spatial layout design of website hierarchical structures, presentation of multimedia contents, and interaction modes, and proposes a VR website design framework based on the fan-shaped layout using the spatial depth cue. Based on this design framework, we built a prototype of the VR version of the website in the context of the online museum website of intangible cultural heritage in Yunnan Province. We provided a reference for spatial layout and visual style design for developing the VR website.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; **Interaction paradigms**; **Virtual reality**;

KEYWORDS

Virtual Reality, Web Design, 3D Layout Framework, web XR

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1 INTRODUCTION

Websites are still the dominant way to visit information on the Internet. The standard devices to visit websites are computers, cell phones, tablets, etc. The rapid development of virtual reality technology in recent years has made many consumer-grade VR devices gradually accepted by the public. VR users can browse websites through the headset's built-in browser or third-party browsers (e.g., Meta Quest Browser and Firefox Reality). VR devices also have the potential to become the next generation of terminal devices for visiting websites [1], so the way of visiting websites in VR is worth investigating. Regarding information presentation, the WIMP interface paradigm usually dominates websites in traditional devices [2,3,4,5]. It is based on the desktop environment, limiting the website's information to two-dimensional space. The VR environment is free from the concept of a desktop, and users will visit information in three-dimensional space. In terms of interaction, traditional devices typically use the keyboard and mouse or touch the screen to interact. In contrast, the VR device's three-dimensional environment allows more diverse interaction modes, such as rays and 3D widgets. This two-dimensional to three-dimensional transformation means the website's layout, media content, and interaction method can be more immersive. It is a challenge for website design.

In recent years, the WebXR API [6] and the A-frame framework [7] based on the Html5 language have supported and facilitated the development of VR versions of websites. Currently, users use VR browsers to visit websites in two ways. Firstly, websites are not developed for Virtual Reality, such websites are presented in a windowed form on the screen in front of the user, but the visual effects of the website content and UI are still consistent with the desktop version. The interaction organization of the website also follows the WIMP interface paradigm, and these desktop-based display methods and keyboard-and-mouse-based interactions for the virtual reality environment are inappropriate [4,5]. It does not reflect the advantages of Virtual Reality with immersion, interactivity, and imagination [4]. Secondly, the website is developed for Virtual Reality. This kind of website provides a virtual reality portal, allowing users to immerse in the virtual scene to visit the website

information. Still, this kind of website is primarily biased towards simulating a particular experience (such as games, museums, etc.). They don't have the ability to visit generic websites. So forming a framework for displaying website information in Virtual Reality is still necessary.

In the current research, Xing et al. have explored a 3D layout framework for static websites, splitting the content, navigation, and scrollbars into components (widgets). It unifies different websites' functions in WebXR by components while combining gesture interaction to provide an immersive WebXR experience [8]. However, the website's content is still presented in the same way as the desktop version. Kharoub et al. have proposed a VR two-layer user interface that allows users to visualize the desktop screen on a 360-degree curved layer by separating the interaction layer from the screen layer [9]. There is research on the presentation of graphical information and information access performance in virtual environments [10, 11, 12, 13]. The above studies have promising implications for improving website experience in VR. Still, most focus on the modular design of specific functions and do not design from the website's information architecture. Regarding information presentation, most studies focus on graphics and text media content while neglecting the integration of immersive media content such as panoramic video, 3D models, and virtual scenes in VR websites.

Therefore, this paper discusses the design elements of VR websites from three aspects and proposes an immersive layout framework for website design in VR. In the spatial layout of the Organizational Structure of the VR website, we use a fan-shaped layout, using the spatial depth cue to arrange the information hierarchy of the website in order from far to near. Regarding the presentation of different multimedia contents, we have carried out the visual design for different 3DUIs. In terms of interaction, we have divided the content area and interaction area so that users can use buttons and ray-casting to interact globally at a close distance. Finally, we developed a prototype of the VR version website in Unreal Engine 4 based on the desktop Yunnan Intangible Cultural Heritage Online Museum website. Two contributions of this study are as follows:

- We proposed a fan-shaped layout framework with information hierarchy and multimedia content integration, distinguishing the content and interaction areas.
- Based on the fan-shaped layout framework, we made a VR prototype for the existing website. We design an immersive layout framework and 3DUI style for the VR website prototype, which provides ideas for developing the VR website.

2 RELATED WORK

2.1 Information Layout in Extended Reality

Xing et al. have explored the way XR devices visit websites. They have designed a three-dimensional layout framework for static websites, splitting the website's content, navigation, and scrollbars into components (widgets) for interaction. The website's main content is presented directly in front of the user as a panel, and other content is shown on the user's oblique above and below as a panel [8]. However, this system's information layout method of the website remains the same as the desktop. In addition, there is also some research on the information layout method in spatial. Kharoub et al. proposed a VR two-layer user interface that allows users to

visualize the desktop screen on a 360-degree curved layer by separating the interaction layer from the screen layer [9]. Barrett Ens et al. proposed the Ethereal Planes framework to assist in managing information through multi-window layouts that map 2D graphic information into 3D mixed reality environments [14]. Barrett M. Ens et al. proposed the Personal Cockpit system, which improves the efficiency of application switching time by 40% [11]. Luo et al. proposed a layout strategy and method for organizing between and within clusters of information windows in an AR environment [10].

Previous research usually places multi-window content in the same spatial location and rarely uses the spatial depth cue to distinguish different content levels. The website's organization structure is roughly a tree-like structure [2, 15], and the information hierarchy is relatively complex. This presentation of Multi-window content with no depth cue is still inapposite for users to perceive the hierarchy of websites in spatial. This study will explore the feasibility of using spatial depth cue to distinguish information hierarchies of websites.

2.2 Presentation of multimedia content on the website

The website can be broken down into specific components we can design, such as page design, navigation, use of multimedia, search design, and URL design [16], of which we focus on multimedia content. The multimedia discussed in this paper refers to a combination of two or more different media types controlled by a computer [17, 18, 22]. In websites, in addition to the standard text, graphics, sound, and video, more immersive forms of multimedia content have emerged in recent years, such as panoramic images [23, 24], 3D models [25, 26], and virtual scenes [27, 28], which are usually interactive. Previous studies have typically examined the presentation of single media content, including the spatial distribution of information [10, 11, 29, 5] and information access performance [12, 13]. Still, most of them are biased towards graphics and text, and there need to be more design solutions and studies that integrate immersive multimedia content into websites.

2.3 Information Interaction in Virtual Environments

D. Mendes et al. summarized how virtual objects interact in 3D space and suggested that the biggest challenge of interacting in 3D space is finding effective metaphors [30]. Several related frameworks have experimented with interaction metaphors in virtual space, such as: interacting with distant objects by extending the distance mapping of fingers or arms [31, 32]; user-centered ray-casting and gaze interaction [33, 34, 5]; miniature world technology [35, 36], where users interact with the miniature virtual world; control interaction through menus and panels [37, 11, 38], etc. Also, try to avoid the "gorilla arm" effect during interactions. [39]. In addition, Argelaguet F et al. summarized the ways to choose virtual objects to confirm [40], among which virtual buttons are standard and effective [41, 42, 43, 44]. The interaction should also generate feedback [45], and shared feedback methods include visual feedback, haptic feedback, auditory feedback, force feedback, and so on [46]. From a usability perspective, interaction selection techniques in virtual

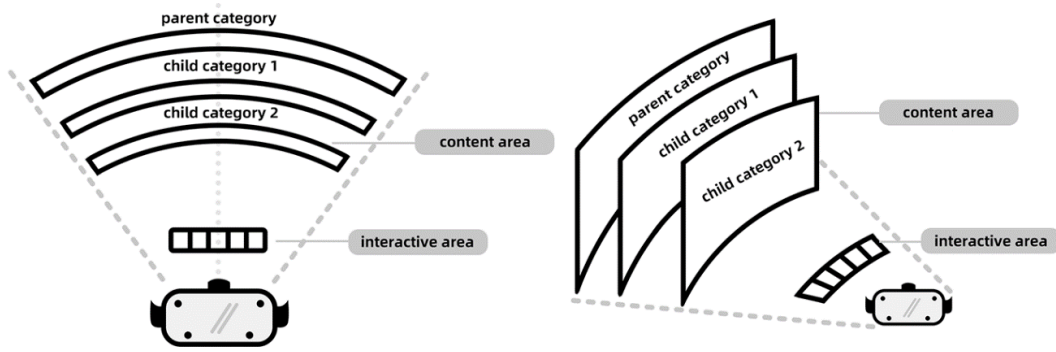


Figure 1: Diagram of the fan-shaped layout framework.

environments must: i) provide fast selection, ii) be accurate and error-proof, iii) be easy to understand and control, iv) produce low levels of fatigue [40].

3 CONCEPT & FRAMEWORK

3.1 Design a Frame

Our system distinguishes between interactive and content areas. The interactive area is set at close distances, and the content area is presented in the middle and far reaches. The interactive area integrates basic operations users require to browse web pages, such as return, confirmation, page-turning, etc. These functions are presented in the form of virtual buttons. The content area presents information, including the website's hierarchy, multimedia content, and navigation. Among them, the website hierarchy is the expression form of the website information organization structure, which usually presents a parent-child relationship. The most general category can be called a parent category, which contains many child categories. The more detailed the classification, the more precise the information presented. Our research aims to design a generic VR website framework that can be applied to websites with a hierarchical structure. Users can distinguish different website hierarchies the visual perception. Based on the principle of field-of-view limitation in VR, the framework principles proposed in this study are as follows (Figure 1):

- The system distinguishes between interactive and content areas. The interactive area is set at close distances, and the content area is presented in the middle and far distances.
- Using the spatial depth cue, we adopt a fan-shaped layout to distinguish different information hierarchies in the VR website.
- In the fan-shaped layout, information is presented in different depths from far to near, and child categories are always closer to the user location than the parent categories.

This chapter will focus on the framework principle from three aspects of VR website design elements: spatial layout design of website hierarchical structures, presentation of multimedia contents, and interaction modes. The framework differentiates VR and

desktop websites to present different visual effects and provide a more immersive website browsing experience.

3.2 Spatial layout design of the hierarchical structure in VR websites

Organizational structure is one of the main components of a website, and one of the typical architectural approaches is "Hierarchical Structures" [15,2]. This structure consists of a parent or global category and branches into child categories below it, which can be broken down into smaller categories until the target object is reached [15,2]. Usually, the information hierarchy of websites [19,20] can be divided into primary menus, secondary menus, tertiary menus, etc. For example, Figure 2 shows a tree structure of a website. Generally speaking, primary menus are usually distributed in the header position of the website and are used to summarize the categories of website content. Secondary menus are a child category of primary menus. And tertiary menus are detailed content under secondary menus. The entire website organization can be integrated into the navigation [2].

In the content area of the VR website system, we rearranged the website's information hierarchy based on the fan-shaped layout framework in a three-dimensions space (Figure 3). The primary menu is a global category of the website content at the farthest distance from the user. The secondary menu is the expanded category at a medium distance. The tertiary menu and additional information are presented as the detailed content closest to the user. These are arranged in an arc with the user at the center. At the same time, we encapsulate the information in each menu into some 3D widget [21, 40] with the 3D widgets at the same menu arranged horizontally and the 3D widgets directly in front of the user being the currently selected content. In addition to the information hierarchy layout, a separate navigation panel displays information about the user's current hierarchy, ensuring that the user does not get lost in the website system [2, 15]. All content area information keeps a distance from the interactive areas.

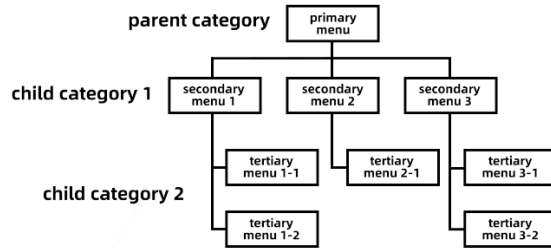


Figure 2: The hierarchical structure of the website.

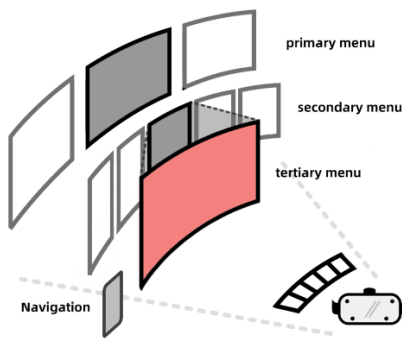


Figure 3: The fan-shaped layout framework.

3.3 Presentation of multimedia content on VR websites

We categorized the multimedia content on the website into five categories: graphics and text, flat video, panoramic image/video, 3D model, and virtual scene. The multimedia content usually presents in the tertiary menu, and graphics and text content are sometimes in the secondary menu. A panel usually supports graphics and text content. Based on the fan-shaped layout framework, we have visually mapped these five multimedia contents in the VR website prototype.

(1) Graphics and Text (Figure 4a): When the user selects to view graphics and text content, that content expands from the parent menu location towards the user, and the information presents in a panel. (2) Flat Video (Figure 4b): The flat video appears like the graphics and text. However, when the video is playing, the buttons in the interactive area will be replaced with video-related functions, such as back, pause, forward, multiply speed, etc. (3) Panoramic image/video (Figure 4c): When selecting the panoramic media, all 3D widgets will switch to unviewable, and all surrounding scenes will be replaced with panoramic media, while the function of the buttons in the interactive area will change. (4) 3D model (Figure 4d): When selecting the 3D model, its graphics and text introduction content on the panel will pop up. At the same time, some miniature 3D models that arc around in front of the user will pop up as a preview effect. Users can switch the preview category of the 3D models by graphics and text panel. When clicking the preview 3D

model, the model will be enlarged, and other preview 3D models will be shrunk to unseen, while the information on the panel will be replaced with the introduction of this model. (5) Virtual Scene (Figure 4e): When selecting a virtual scene, the panel will present the icon that can entrance to enter the virtual scene. After entering, all 3D widgets and buttons will be switched to unseen, and users can have an immersive roaming experience inside the virtual scene.

3.4 Interaction design in VR websites

Users can browse the website through the VR's built-in browser or a third-party browser. Take Meta Quest Browser to enter Meta's website as an example: the browser presents to the user a window panel of customizable size, with a fixed toolbar at the window's top that includes functions such as refresh, back, URL search box, bookmarks, and menu, etc. Below the toolbar is the website's content panel. All icons and content interact with ray-casting.

We divided the content and interaction areas to build more natural interaction on the VR website (Figure 1). The content area is presented in the middle and far distances from the user, where the user can visit the information on the website. The interaction area is set within reach of the user's hands to realize the global interaction at a close distance. The metaphor for the interaction area is the button, and these virtual buttons also wrap around the user's front in an arc shape. We selected five functions for buttons: back, previous, confirm, next, and home (the functions can be customized), and the user can control the virtual hand to press the button down through the controller to trigger the interaction of the corresponding function (Figure 5-a). The button function can be replaced when specific content is displayed, such as playing a video. Ray-casting interaction (Figure 5-b) was added in situations requiring accurate selection^[30], such as switching categories, selecting a specific preview model, etc. Ray-casting typically controls content-related interactions, while virtual buttons typically control non-content-related interactions. To avoid the adverse effects of object occlusion^[40], we staggered the placement of all interactable elements on the visual angle. Both button and ray-casting interactions have displacement animations or highlighting to provide visual feedback, while the controller generates vibration feedback to improve the interaction experience.

4 IMPLEMENTATION

Based on the online website of Yunnan Intangible Cultural Heritage(ICH) on PC/mobile, we built a VR version prototype of the website by using the fan-shaped layout framework.

The ICH online website has a three-level menu structure. The primary menu includes "Home," "ICH Experience," "ICH Projects," and "ICH Inheritors." where "ICH Experience" has five secondary menus. One of them is "ICH Virtual Museum." Under "ICH Virtual Museum," the tertiary menu includes information on 26 specific museums. Users can visit five multimedia contents on the website: graphics and text, flat videos, panoramic images and videos, 3D models, and virtual scenes. Among them, PC users can use the mouse to click to view graphic, text, and video information or use the mouse to drag and drop to view panoramic media and 3D models. Double-click on the scene ground, or use the keyboard

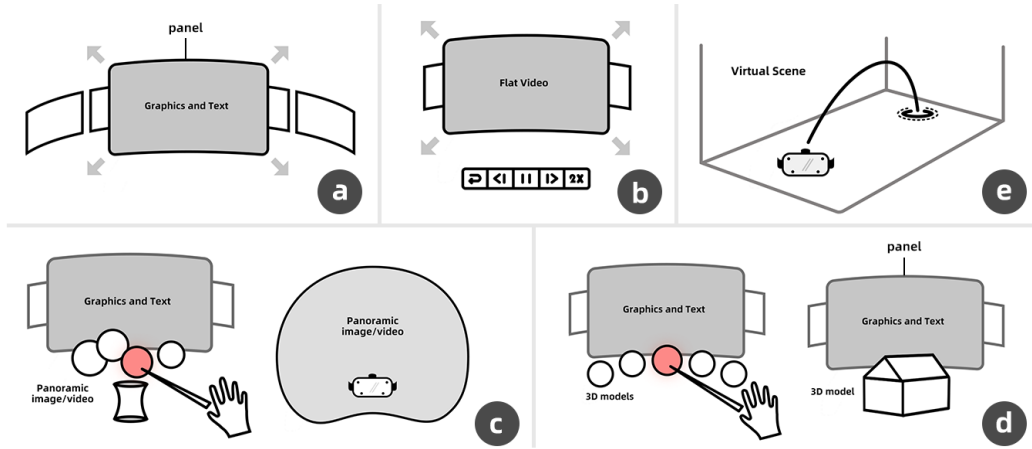


Figure 4: Five categories of multimedia presentation.

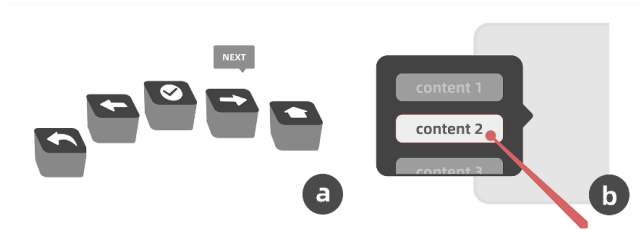


Figure 5: (a) 3D buttons in the interaction area; (b) Ray-casting interaction.

arrow keys to roam in the virtual scene. Mobile users can browse with a touch instead of a mouse function.

In the VR prototype, we used the spatial depth cue to distinguish the content and interaction areas so that all modules are in a fan-shaped layout framework. We designed the UI for multimedia content differently from the desktop. When previewing panorama and 3D data, we used spherical wrapping for visual presentation. Figure 6 shows the visual representation of each level menu, and Figure 7 shows the visual representation of different multimedia contents.

At the same time, we clarify the way to select and confirm information. When entering the system, the user is in the home position of the website, and three categories of the primary menu are displayed directly in front. The user can press the left and right arrow buttons in the interactive area to switch the selected category horizontally. The user is in front of the currently selected category, and the selected category is highlighted. Suppose the user wants to choose the "ICH Experience" category in the primary menu, he can press the confirm button when the category is highlighted. At this moment, the primary menu will be shrunk, and the secondary menu will be expanded to a location closer to the user. When the video is playing, the buttons in the interactive area change to video-related functions. At any time the user wants to view the content of the previous level menu, they can press the "Back" button in front of them. Ray-casting interaction was added in situations requiring

accurate selection, such as selecting a virtual museum and specific nationality categories. In the virtual museum scene, the user can use the joystick to control teleportation.

5 CONCLUSION & FUTURE WORK

This study aims to improve the experience of browsing websites in VR and proposes an immersive VR website design framework that includes multimedia content. This paper explores the current status and problems of visiting websites with VR devices. Moreover, combine the shortcomings in other studies to present different information hierarchies in the website by using the spatial depth cue, and optimizing the visual and interactive experience based on multimedia content. Therefore, we propose a VR website design framework based on a fan-shaped layout. We analyze the design elements from three aspects: spatial layout design of website hierarchical structures, presentation of multimedia contents, and interaction modes. Based on this design framework, we built a prototype of the VR version of the website in the context of the online museum website of intangible cultural heritage in Yunnan Province.

This is still a work in progress, and we will evaluate and validate the usability of the framework and prototype in the future. In addition, many switching animations have been added to the prototype, but the impact of the length of the switching animations on the user experience has yet to be considered. Meanwhile, the interaction method of button and ray-casting fusion and the presentation of 3DUI also need to be analyzed and improved by user feedback. In the future, we will continue to improve the design framework and optimize the interactive prototype. We will research the universality of the fan-shaped layout framework aimed at mainstream websites to bring a more immersive VR website browsing experience.

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Figure 6: The visual presentation of each menu.



Figure 7: The visual presentation of different multimedia contents. (a) Graphics and Text; (b) Flat Video; (c) 3D model; (d) Panoramic image/video; (e) Virtual Scene.

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